

ADRIAN E. BAYER

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 Boston, USA

ACADEMIC APPOINTMENTS

Massachusetts Institute of Technology, USA IAIFI Fellow	2026–
Harvard–Smithsonian Center for Astrophysics, USA IAIFI Fellow	2026–
Perimeter Institute, Canada Fellow	2026–
Flatiron Institute, Simons Foundation, USA Flatiron Research Fellow	2023–2026
Princeton University, USA Postdoctoral Researcher	2023–2026
The University of Tokyo, Japan Visiting Researcher (3 months)	2022
Massachusetts Institute of Technology, USA Undergraduate Researcher (2 months)	2016 Adviser: Lindley Winslow
Imperial College London, UK Undergraduate Researcher (3 months)	2015 Adviser: Henrique Araújo

EDUCATION

University of California, Berkeley, USA Ph.D. Physics	2018–2023 Thesis adviser: Uroš Seljak
University of Cambridge, UK Master of Advanced Study, Mathematics	2017–2018
Imperial College London, UK MSci Physics with Theoretical Physics Graduated top of the cohort (approx. 250 students)	2013–2017 Thesis adviser: Fay Dowker

HONORS AND AWARDS

Blavatnik Regional Award for Young Scientists in Physical Sciences and Engineering (\$10,000), Blavatnik Family Foundation and The New York Academy of Sciences, 2026

Outstanding Graduate Student Instructor Award, University of California, Berkeley, 2022

Berkeley Distinguished Graduate Fellows Video Prize (\$1,000 grant), University of California, Berkeley, 2019

The Berkeley Fellowship, University of California, Berkeley, 2018

Abdus Salam Undergraduate Prize, Imperial College London, 2017

Governors' MSci Prize in Physics, Imperial College London, 2017

Ken Allen Prize, Imperial College London, 2016

Winton Capital Prize for Outstanding Performance in Second Year Physics, Imperial College London, 2015

EPSRC Summer Vacation Bursary (£2,200 grant), Engineering and Physical Sciences Research Council, 2015

COLLABORATIONS AND LEADERSHIP

Simons Observatory

Leader of Foreground / Sky Modeling Working Group
Leader of AI Initiative
Member of Digital Assets & AI Governance Committee

DESI

Member

LSST DESC

Member

Learning the Universe

Member

ACADEMIC SERVICE

SOC+LOC of **CMB Summer School** at Flatiron Institute (2026)
SOC of **UrošFest – From Pen and Paper to Petaflops: Cosmology’s 30-Year Transformation** (2026)
SOC of **MOCK NYC** at New York (2026)
Organizer of **Astrophysics x ML Meeting** at Flatiron Institute (2025-)
Organizer of **Dark Cosmos Seminar** at Princeton University (2025)
Organizer of **Cosmology Lunch Meeting** at Princeton University (2024-)
Organizer of **Cosmology and ML Meeting** at Simons Foundation (2024-)
Convener of **Numerical Cosmology and Artificial Intelligence** session at COSMO’24 [Japan] (2024)
Organizer of **Debating the potential of machine learning for astronomical surveys (#2)** conference at IAP/CCA (2023)

TEACHING

Machine Learning and Simulations Lecturer, CMB Summer School, Flatiron Institute, USA	2026
Statistical Modeling and Introduction to Machine Learning Lecturer, AstroAI Asian Network Summer School, Seoul, Korea	2025
Astronomy Lecturer, East Jersey State Prison, Rahway, USA	2024
Bayesian Data Analysis and Machine Learning for Physical Sciences Graduate Student Instructor, UC Berkeley, USA	2021
Python for Physics Teaching Assistant, Imperial College London, UK	2016

MENTORING

Miranda McCarthy (Masters Student), City University of New York (2025–2026)
Career mentor

Carmen Émbil Villagr  (Graduate Student), University of Cambridge (2025–)
Project: Nonlinear information in the tSZ

Liam Parker (Graduate Student), UC Berkeley (2024–)

Project: Field-Level Inference using CNNs (co-supervised with Uroš Seljak)

Paper: [arXiv: 2504.01092](#) ([astro-ph.CO](#))

Shaunak Padhyegurjar (Undergraduate Student), IISER Bhopal, India (2024–)

Project: Primordial non-Gaussianity with Voids

Veena Krishnaraj (Undergraduate Student), Princeton University (2024–5)

Project: Beyond the Standard Model with Machine Learning

Paper: [arXiv: 2510.19168](#) ([astro-ph.CO](#))

James Robinson (Undergraduate Student), Princeton University (2024–)

Project: Fast Algorithms for Computing the Kinetic Sunyaev-Zeldovich Effect

Arnab Lahiry (Undergraduate Student), Indian Institute of Science Education and Research, Tirupati (2024–5)

Project: Interpreting the Information in the Cosmic Web (co-supervised with Francisco Villaescusa-Navarro)

Paper: [arXiv: 2504.17839](#) ([astro-ph.CO](#))

Chenze Dong (Graduate Student), University of Tokyo (2024–)

Project: Field-Level Inference for Galaxies (co-supervised with Ben Horowitz and KG Lee)

Paper: [arXiv: 2604.26823](#) ([astro-ph.CO](#))

Akira Tokiwa (Graduate Student), University of Tokyo (2023–5)

Project: Impact of Box Size for Weak Lensing (co-supervised with Jia Liu and Masahiro Takada)

Paper: [arXiv: 2511.20423](#) ([astro-ph.CO](#))

James Sunseri (Graduate Student), Princeton University (2023–5)

Project: Information Content of the Cosmic Web (co-supervised with Jia Liu)

Paper: [arXiv: 2503.11778](#) ([astro-ph.CO](#))

Yici Zhong (Graduate Student), University of Tokyo (2022–4)

Project: HalfDome Cosmological Simulations for Stage IV Surveys (co-supervised with Jia Liu)

Paper: [arXiv: 2407.17462](#) ([astro-ph.CO](#))

Malika Golshan (Undergraduate Student), UC Berkeley (2022–4)

Project: Interpreting what AI learns about neutrino physics in cosmological simulations

Paper: [arXiv: 2410.00914](#) ([astro-ph.CO](#))

Jakob Robnik (Graduate Student), UC Berkeley (2021–4)

Project: Data-Driven Noise Models and Look-Elsewhere Effect (co-supervised with Uroš Seljak)

Paper: [arXiv: 2407.17565](#) ([astro-ph.GA](#))

REFEREEING

Journals: Astronomy & Astrophysics (A&A), The Astrophysical Journal (ApJ), Astrophysical Journal Letters (ApJL), International Conference on Machine Learning (ICML), Journal of Cosmology and Astroparticle Physics (JCAP), Monthly Notices of the Royal Astronomical Society (MNRAS), Neural Information Processing Systems (NeurIPS), Physical Review D (PRD), Physical Review Letters (PRL), Physical Review X (PRX)

Grants: ERC, IVADO

OUTREACH

Invited scientist (“ask a scientist”) @ NASA Moon Tree Launch Party, Madison Square Park Conservancy, New York NY, USA (2026)

Short film co-director, Concrete Cosmos, Labocine—Simons Foundation Symbiosis program; premiered at DCTV, New York NY USA (2026)

Astronomy on Tap @ Jersey Shore, Organizer (Summer 2025)

Astronomy on Tap @ Trenton, Host and Organizer (2023-2025)

Presenter at City College of New York STEM Career Fair (2024)

Media Representative at the Flatiron-Nomad Partnership and Simons Foundation Public Eclipse Party (2024)

Berkeley Compass Mentor (2022)

Adopt-a-Physicist Mentor (2020)

OUTREACH TALKS

Reflections on the Moon: *a conversation with Tracy Scott and Denise Hill* December 2025
Simons Foundation, New York NY, USA

Our Learning Universe: AI Meets the Cosmos October 2025
Flatiron Institute, New York NY, USA

Neutrino Mass from Cosmology: Measuring the Mass of a Needle in a Haystack August 2024
New Jersey State Museum, Planetarium, Trenton NJ, USA

Neutrino Mass from Cosmology: Measuring the Mass of a Needle in a Haystack November 2023
Astronomy on Tap, Trenton NJ, USA

SELECTED ACADEMIC TALKS

Space Telescope Science Institute, Baltimore, MD, USA August 2026
Surveying the Universe in 4D Workshop
“Optimally Surveying the Universe in 4D: Machine Learning and Field-Level Inference”

Universidade da Coruña, A Coruña, Spain August 2026
New Frontiers in Cosmology: Cosmology under Totality
“Field-level inference for multi-probe cosmology” & Panelist for “AI in Cosmology Discussion”

University of Toronto, Toronto, Canada July 2026
Simons Observatory F2F Meeting
“Machine Learning and Simulations”

University of Virginia, Charlottesville, VA, USA July 2026
2nd Annual NSF–Simons CosmicAI Science Conference
“Simulation-based inference for multi-probe cosmology”

Perimeter Institute for Theoretical Physics, Waterloo, Canada June 2026
Advancing Field-Level and Simulation-Based Inference for Cosmology Workshop
“FLISBI for LSSCMB” & panelist for AI discussion

UrošFest, San Francisco, CA, USA May 2026
From Pen and Paper to Petaflops: Cosmology’s 30-Year Transformation
“Reconstructing Grad School with Uros-Level Inference” & chair and panelist for AI Discussion

60th Rencontres de Moriond, La Thuile, Italy March 2026
Cosmology Conference
“Extracting optimal and robust information from cosmological surveys with simulation-based inference”

Kavli Institute for Theoretical Physics, UC Santa Barbara, CA, USA February 2026
Exploring New Boundaries in Cosmology and Astrophysics: CMB and LSS Workshop
“Simulations for ML Applications: Prospects and Challenges for Multi-Probe Simulation-Based Inference”

University of Texas, Austin, TX, USA Weinberg Seminar “Unifying the Sky: Field-Level Inference for Multi-Probe Cosmology”	February 2026
University of Texas, Austin, TX, USA Physics Colloquium “Why a tiny neutrino particle inspired me to simulate and reconstruct the entire Universe”	February 2026
Kavli Institute for Particle Astrophysics & Cosmology, Stanford University, CA, USA KIPAC Seminar “Field-Level Inference in the Multimodal Cosmos: Scaling Scientific Discovery Across Fields with AI”	February 2026
Kavli Institute for Theoretical Physics, UC Santa Barbara, CA, USA New Synergies in Multi-Probe Cosmology Conference “A Review of Machine Learning and AI for Cosmology and Astrophysics”	February 2026
Institute of Astronomy and Astrophysics, Academia Sinica, Taipei, Taiwan Reconstructing the Universe Workshop “Field-Level BAO Reconstruction and beyond”	December 2025
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Cosmology X Data Science Meeting “Why a tiny neutrino particle inspired me to simulate and reconstruct the entire Universe”	November 2025
Rutgers University, New Brunswick NJ, USA New High Energy Theory Center Seminar “Why a tiny neutrino particle inspired me to simulate and reconstruct the entire Universe”	November 2025
University of Waterloo, Waterloo, Canada WCA Astroseminar “What’s the Likelihood? Field-Level Cosmology x Detecting Signals in Large and Noisy Places”	October 2025
Perimeter Institute, Waterloo, Canada Cosmology Meeting “Why a tiny neutrino particle inspired me to simulate and reconstruct the entire Universe”	October 2025
ETH Zürich, Zürich, Switzerland CosmoClub Seminar “Why a tiny neutrino particle inspired me to simulate and reconstruct the entire Universe”	September 2025
Erwin Schrödinger International Institute, Vienna, Austria Putting the Cosmic Large-scale Structure on the Map: Theory Meets Numerics “Field-Level BAO Reconstruction and Beyond”	September 2025
Max Planck Institute for Astrophysics, Munich, Germany Cosmology Meeting “Field-Level BAO Reconstruction and Beyond”	August 2025
University of Cambridge (Kavli Institute for Cosmology), Cambridge, UK CMB/LSS Meeting “Field-Level BAO Reconstruction and Beyond”	August 2025
Centro de Ciencias de Benasque Pedro Pascual, Benasque, Spain Understanding Cosmological Observations Workshop “Field-Level BAO Reconstruction”	July 2025
University of Manchester, UK Simons Observatory Collaboration Meeting “Sky Modeling”	July 2025

Sexten Center for Astrophysics, Sexten, Italy New Strategies For Extracting Cosmology From Future Galaxy Surveys Workshop – 3rd Edition “Field-Level BAO Reconstruction”	July 2025
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Exoplanet Meeting “Detecting astronomical signals in large and noisy spaces”	June 2025
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Cosmology X Data Science Meeting “What’s the Likelihood? Field-Level Cosmology x Detecting Signals in Large and Noisy Places”	May 2025
Center for Astrophysics, Harvard University, Cambridge MA, USA AstroAI Seminar “Why a tiny neutrino particle inspired me to reconstruct the initial conditions of the Universe”	April 2025
Center for Astrophysics, Harvard University, Cambridge MA, USA ITC Luncheon Talk “Field-Level Reconstruction of the Cosmological Initial Conditions”	April 2025
Massachusetts Institute of Technology, Cambridge MA, USA TESS Science Talk “Detecting astronomical signals in large and noisy spaces”	April 2025
University of Pennsylvania, Philadelphia PA, USA Astro Seminar “Why a tiny neutrino particle inspired me to simulate and reconstruct the entire Universe”	April 2025
Tokyo University, Tokyo, Japan Astronomy Seminar “Towards an Optimal Cosmological Detection of Neutrino Mass with Field-Level Inference and Joint Analyses”	November 2024
Kyoto University, Kyoto, Japan COSMO’24 “Introduction to Numerical Cosmology and Artificial Intelligence”	October 2024
University of Chicago, Chicago IL, USA Simons Observatory Collaboration Meeting “CMB x LSS with the HalfDome Simulations”	July 2024
Mediterranean Institute for Life Sciences, Split, Croatia Cosmology in the Adriatic – From PT to AI “Cosmology in the Adriatic with Adrian: from field-level inference to joint analyses”	July 2024
Università degli Studi di Catania - Dipartimento di Fisica e Astronomia, Catania, Italy International Conference on Machine Learning for Astrophysics – 2nd Edition “Extracting optimal information from cosmological surveys with field-level inference and joint analyses”	July 2024
Sexten Center for Astrophysics, Sexten, Italy New Strategies For Extracting Cosmology From Future Galaxy Surveys Workshop – 2nd Edition “The HalfDome CMB x LSS Simulations”	July 2024
Aspen Center for Physics, Aspen CO, USA Fundamental Physics in the Era of Big Data and Machine Learning “Physics-based sampling”	June 2024
Grand Arsenal, Chania, Greece COSMO21: Statistical Challenges in 21st Century Cosmology “Towards an Optimal Cosmological Detection of Neutrino Mass with Joint Analyses and Field-Level Inference”	May 2024

Stanford University, Stanford CA, USA Cosmology Seminar “Towards an Optimal Cosmological Detection of Neutrino Mass with Field-Level Inference”	April 2024
Yale University, New Haven CT, USA Cosmology Seminar “Towards an Optimal Cosmological Detection of Neutrino Mass with Field-Level Inference”	April 2024
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Cosmology X Data Science Meeting “The HalfDome CMB x LSS Simulations”	March 2024
The Center for Cosmology and Particle Physics, New York University, NY, USA Astrophysics and Relativity Seminar “Towards an Optimal Cosmological Detection of Neutrino Mass with Field-Level Inference”	March 2024
Center for Data Driven Discovery (CD3), Kavli IPMU, University of Tokyo, Japan The CD3 x Simons Foundation workshop: AI-driven discovery in physics and astrophysics “Cosmological Field-Level Inference with Microcanonical Langevin Monte Carlo”	January 2024
High Energy Accelerator Research Organization (KEK), Tsukuba, Japan ML at HEP workshop “Extracting optimal information from upcoming cosmological surveys”	January 2024
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Debating the potential of machine learning for astronomical surveys (#2) – IAP/CCA Conference “Cosmological Field-Level Inference with Microcanonical Langevin Monte Carlo”	November 2023
Imperial College London, London, UK Seminar “Towards an Optimal Cosmological Detection of Neutrino Mass”	November 2023
Monte Verità, Ascona, Switzerland Hamers & Nails, Frontiers in Machine Learning in Cosmology, Astro & Particle Physics “Cosmological Field-Level Inference with Microcanonical Langevin Monte Carlo”	October 2023
Hawaii Convention Center, Honolulu HI, USA International Conference on Machine Learning (ICML), Workshop on Machine Learning for Astrophysics “Field-Level Inference with Microcanonical Langevin Monte Carlo”	July 2023
Sixten Center for Astrophysics, Sixten, Italy New Strategies For Extracting Cosmology From Future Galaxy Surveys Workshop “Field-Level Inference with Microcanonical Hamiltonian Monte Carlo”	July 2023
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Cosmic Connections Symposium “Field-Level Inference with Microcanonical Hamiltonian Monte Carlo”	May 2023
Stanford University, Stanford CA, USA (zoom) LSST Higher-Order Statistics Meeting “Super-Sample Covariance of Higher-Order Statistics”	January 2023
Institute for Advanced Studies, Princeton NJ, USA Cosmology Lunch “Towards Optimal Measurement of the Neutrino Mass using Large-Scale Structure”	December 2022
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Cosmology X Data Science Meeting “Towards Optimal Measurement of the Neutrino Mass using Large-Scale Structure”	December 2022

University of Pennsylvania, Philadelphia PA, USA CMB Meeting “Towards Optimal Measurement of the Neutrino Mass using Large-Scale Structure”	December 2022
Université de Montréal, Montréal, Canada Astromerique Speaker Series “Massive Neutrino Information in Large-Scale Structure and Field-Level Inference”	November 2022
Vipolže, Slovenia Berkeley Center for Cosmological Physics Summer Workshop “Massive Neutrino Information in Large-Scale Structure and Field-Level Inference”	July 2022
The University of Tokyo (Hongo Campus), Tokyo, Japan GPU Workshop “Cosmological simulations on GPU with tensorflow”	May 2022
The University of Tokyo (Kavli IPMU), Kashiwanoha, Japan APEC Seminar “Towards detecting neutrino mass using non-linear cosmic structure”	April 2022
Kyoto University (Yukawa Institute for Theoretical Physics), Kyoto, Japan Cosmology with Weak Lensing: Beyond the 2-pt Statistics “Detecting neutrino mass using nonlinear cosmic structure”	April 2022
Institut d’Astrophysique de Paris, Paris, France Journal Club – Univers “Detecting neutrino mass using nonlinear cosmic structure”	February 2022
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Learn the Universe “The Look-Elsewhere Effect”	August 2021
Pennsylvania State University, State College PA, USA Statistical Challenges in Modern Astronomy VII “The Look-Elsewhere Effect from a Unified Bayesian and Frequentist Perspective”	June 2021
University of Cambridge (Kavli Institute for Cosmology), Cambrigde, UK KICC 10th Anniversary Symposium “Look Elsewhere” (poster)	September 2019

BIBLIOGRAPHY

PRIMARY AUTHOR:

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2. **Bayer, A. E.**, L. Parker, D. Valcin, S.-F. Chen, C. Modi, and U. Seljak. “Field-Level Inference from Galaxies: BAO Reconstruction”. In: (Mar. 2026). arXiv: [2603.15732 \[astro-ph.CO\]](#)
3. Tokiwa, A., **A. E. Bayer**, J. Armijo, J. Liu, R. Terasawa, L. Thiele, M. Alvarez, L. Blot, and M. Takada. “Impact of Simulation Box Size for Weak Lensing: Replication and Super-Sample Effects”. In: (Nov. 2025). arXiv: [2511.20423 \[astro-ph.CO\]](#)
4. Horowitz, B. and **A. E. Bayer**. “jFoF: GPU Cluster Finding with Gradient Propagation”. In: (Oct. 2025). arXiv: [2510.26851 \[astro-ph.IM\]](#)
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6. Thiele, L., **A. E. Bayer**, and N. Takeishi. “Simulation-Efficient Cosmological Inference with Multi-Fidelity SBI”. in: (July 2025). arXiv: [2507.00514 \[astro-ph.CO\]](#)
7. **Bayer, A. E.**, F. Villaescusa-Navarro, S. Sharief, R. Teyssier, L. H. Garrison, L. Perreault-Levasseur, G. L. Bryan, M. Gatti, and E. Visbal. “Field-level Comparison and Robustness Analysis of Cosmological N-body Simulations”. In: *ApJ* 989.2 (2025), p. 207. DOI: [10.3847/1538-4357/ade4e](#). arXiv: [2505.13620 \[astro-ph.CO\]](#)
8. Lahiry, A., **A. E. Bayer**, and F. Villaescusa-Navarro. “Interpreting Cosmological Information from Neural Networks in the Hydrodynamic Universe”. In: *The Astrophysical Journal* 994.1 (Apr. 2025), p. 129. DOI: [10.3847/1538-4357/ae0b5a](#). arXiv: [2504.17839 \[astro-ph.CO\]](#)
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12. Golshan, M. and **A. E. Bayer**. “Massive ν s through the CNN lens: interpreting the field-level neutrino mass information in weak lensing”. In: *JCAP* 05 (2025), p. 024. DOI: [10.1088/1475-7516/2025/05/024](#). arXiv: [2410.00914 \[astro-ph.CO\]](#)
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19. **Bayer, A. E.**, A. Banerjee, and U. Seljak. “Beware of fake ν ’s: The effect of massive neutrinos on the nonlinear evolution of cosmic structure”. In: *Phys. Rev. D* 105.12, 123510 (June 2022), p. 123510. DOI: [10.1103/PhysRevD.105.123510](#). arXiv: [2108.04215 \[astro-ph.CO\]](#)
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24 (Sept. 2021), p. 24. DOI: [10.3847/1538-4357/ac0e91](https://doi.org/10.3847/1538-4357/ac0e91). arXiv: [2102.05049](https://arxiv.org/abs/2102.05049) [[astro-ph.CO](#)]

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23. **Bayer, A. E.** and U. Seljak. “The look-elsewhere effect from a unified Bayesian and frequentist perspective”. In: *J. Cosmology Astropart. Phys.* 2020.10, 009 (Oct. 2020), p. 009. DOI: [10.1088/1475-7516/2020/10/009](https://doi.org/10.1088/1475-7516/2020/10/009). arXiv: [2007.13821](https://arxiv.org/abs/2007.13821) [[physics.data-an](#)]

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